

publishing in top journals. Thus, productivity and channel quality do not act as substitutes but reinforce each other.

Among the control variables, the overall level of cooperation (through publications) and academic age show a positive effect, while gender does not significantly affect the probability of belonging to the publishing elite after taking into account other factors. The neutral role of gender is especially interesting in the context of numerous studies linking high research productivity with being male (Abramo et al., 2009; Kwiek, 2016) – but not surprising in Poland (Kwiek, 2018a). The diagnostic characteristics of the model confirm its stability, good identifiability, and high ability to describe the mechanisms of selection to the publishing elite.

The aim of this study was to determine whether the advantage of top performers over other scientists lies solely in their higher publication intensity, or whether it is qualitative in nature and manifests itself primarily in selective access to prestigious journals. The results clearly indicate the latter. In all the disciplines and periods examined, top performers achieve an incomparably higher share of publications in prestigious journals (90–99 Scopus journal percentile ranks), while publishing in the lower segments of the journal hierarchy does not differentiate the two classes to a similar extent.

The key finding is therefore that the stratification between top performers and other authors is not evenly distributed throughout the entire publication structure but concentrated in its narrow, highest segment. The differences between the two classes of scientists are therefore not purely quantitative (that is, top performers having more publications “everywhere”) but qualitative: top performers are overrepresented precisely where rejection rates are highest and access to publication channels is most limited – in top journals.

These results allow us to reinterpret the concept of publishing elite. Traditional approaches to top performers, based on the number of publications or their total volume, suggested the existence of a continuous productivity gradient. Our analyses show, in contrast, that the most important division is at the level of the quality of the publication channel and not the intensity of knowledge production itself. Elite publishing status in science thus turns out to be a discrete and threshold phenomenon: what matters is not whether a scientist publishes a lot but whether they publish in the most prestigious journals. Top performers (the upper 10%) tend to publish in top journals (the upper 10%), which makes them the global publishing elite.

This picture is consistent with the classical theories of cumulative advantage and the Matthew effect. At the same time, it shifts the focus to institutional selection mechanisms. Prestigious journals function not only as places of knowledge transmission but also as outlets that provide international visibility, citations, and further publication opportunities. Top journals are not a neutral “mirror” of research quality but an active element of the stratified structure of the global science system.

We empirically confirm that productivity and the prestige of publication channels are complementary rather than supplementary. Our models show that publishing intensity increases the likelihood of belonging to the publishing elite primarily when it is coupled with publishing in top journals. The sheer number of publications, without the quality component, has a much weaker selection power.

This finding has an important implication suggesting selection mechanisms at work in the science system: access to prestigious publication channels powerfully differentiates positions taken by the publishing research population. The publishing elite do not seem to emerge from diligence and effort only but also from successful entries into selective global publication circuits.

Our cross-disciplinary analyses show that the stratification mechanisms described above are permanent and structural. Although the absolute level of participation in top journals varies between disciplines – reflecting different publication cultures and journal structures – the relative advantage of top performers over other scientists remains consistent across disciplines. Similarly, despite the overall increase in publications and the expansion of the Polish scientific system over time, inequalities in access to top journals are not diminishing. This means that the patterns observed are not temporary; on the contrary, with the expansion of the Polish science system, the importance of selective mechanisms may be increasing as growing competition for visibility and prestige enhances the role of publication channels as selection tools.

Our findings have direct implications for research on academic careers. If access to top journals is highly concentrated in the hands of top performers and stable over time, it should be treated as a key mechanism of career gatekeeping rather than as merely a derivative of high productivity. Publishing in prestigious journals increases visibility, strengthens reputation, and influences institutional decisions regarding tenure, promotions, and grants. In this sense, the study shifts the emphasis from the question of “who publishes more” to the question of “who gains access to prestigious channels – and with what regularity.” The answer to the latter question is crucial to understanding why inequalities in science are so persistent and how publishing elites reproduce themselves.

Like any bibliometric study, this one has limitations. First, the prestige of journals was operationalized using Scopus percentile ranks, which do not exhaust all dimensions of research quality and significance. The major advantage of the Scopus percentile ranks system (CiteScore) is that it is citation-based and covers about 30,000 active journals (2026). Second, our analysis focused on article publications and did not cover other forms of knowledge production, which may have important roles in some disciplines. For this reason, we included only STEMM disciplines, as in Poland, outputs in the humanities and social sciences are still predominantly published in Polish. Third, although our results suggest strong career implications, bibliometric data alone do not allow for direct observation of institutional decisions such as promotions or grant awards. Our dataset does not cover critical variables such as departmental “climates” and “personalities” or other department-level data (research intensity, access to funding, staff structure, international collaboration, gender parity, students/staff ratio, etc.; Fox & Mohapatra, 2007; Leisyte & Dee, 2012; Tang & Horta, 2023; Turner & Mairesse, 2005). However, these limitations do not undermine the main conclusions of the study but point to the need for research combining bibliometric data with administrative (and qualitative) data, especially in the context of full academic career trajectories.

In summary, we have shown that the emergence of publishing elites in science is not a simple effect of linear accumulations of publications. The advantage of top performers over other scientists is revealed primarily in selective access to prestigious journals (rather than in a uniform increase in publishing activity in the entire hierarchy of publishing channels). Top journals emerge as playing a key stratification role. From a broader perspective, our results suggest that the contemporary science system is increasingly organized around qualitative thresholds of access rather than continuous gradients of productivity. Understanding who crosses these thresholds and why is essential for a reliable analysis of inequality in science. The micro-panel dataset used in the study (scientist $\times$ period $\times$ discipline), covering a long time horizon and containing detailed information on journal prestige and the publication intensity of individuals, enables a transition from static class comparisons to analyses of dynamic selection processes.

Finally, there are a number of directions for future research. We could model access to the segment of the most prestigious journals (90–99 Scopus percentile ranks) as events in time. It is possible to define the time from the first publication to the first publication in this segment and estimate survival models taking into account time-varying variables such as the cumulative number of publications, the

average prestige of previous journals, co-authorship intensity, or the share of international collaboration. This would allow us to identify qualitative changes in publication trajectories across genders and disciplines. We could also separate *entering* from *remaining* in the publishing elite. Scientists could be assigned to states of having no publications in top journals, having an episodic presence, and maintaining a regular presence in top journals. The probabilities of transitions between these states could then be analyzed to assess the strength of elite reproduction by gender and discipline.

Another area for research could be the identification of heterogeneity within the top performer class. The data allow us to distinguish between trajectories (e.g., latent class analysis) using the number of publications, the share of publications in top journals, the variability of prestige, and the rate of change in these indicators, which make it possible to identify different strategies of entering the class of top performers. An important extension of this research would also be to take into account the nonlinear effects of academic age and cohorts entering the publication system, e.g., through interactions between journal prestige and academic age in logistic regression models. This would make it possible to determine whether the importance of top journals is greater in the early stages of an academic career and whether the window of entry to the publishing elite narrows with age.

Our study clearly shows that it is top journals (the top 10%) that make the difference for membership in the top performer class (top 10%). In the Polish context, top journals tend to be open to top performers – and they tend to be relatively, sometimes fully, closed to remaining scientists. However, on a cautionary note, the top 10% of journals will always be only 10% of them. It is impossible to expect masses of scientists to be publishing in them, as their principal feature is the scarcity of publishing space.

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Author contributions

Marek Kwiek: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Writing—original draft, Writing—review & editing. Wojciech Roszka: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing.

Competing interests

The authors have no competing interests.

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Data availability

We used data from Scopus, a proprietary scientometric database. For legal reasons, data from Scopus received through collaboration with the ICSR Lab (Elsevier) cannot be made openly available.

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